arises due to the edges between θ, $\mathbf{z}$, and $\mathbf{w}$. By dropping these edges and the $\mathbf{w}$ nodes, and endowing the resulting simplified graphical model with free variational parameters, we obtain a family of distributions on the latent variables. This family is characterized by the following variational distribution:

$$q(\theta, \mathbf{z} \mid \gamma, \phi) = q(\theta \mid \gamma) \prod_{n=1}^{N} q(z_n \mid \phi_n), \tag{4}$$

where the Dirichlet parameter γ and the multinomial parameters $(\phi_1, \dots, \phi_N)$ are the free variational parameters.

Having specified a simplified family of probability distributions, the next step is to set up an optimization problem that determines the values of the variational parameters γ and ϕ. As we show in Appendix A, the desideratum of finding a tight lower bound on the log likelihood translates directly into the following optimization problem:

$$(\gamma^*, \phi^*) = \arg\min_{(\gamma,\phi)} \mathrm{D}(q(\theta, \mathbf{z} \mid \gamma, \phi) \parallel p(\theta, \mathbf{z} \mid \mathbf{w}, \alpha, \beta)). \tag{5}$$

Thus the optimizing values of the variational parameters are found by minimizing the Kullback-Leibler (KL) divergence between the variational distribution and the true posterior $p(\theta, \mathbf{z} \mid \mathbf{w}, \alpha, \beta)$. This minimization can be achieved via an iterative fixed-point method. In particular, we show in Appendix A.3 that by computing the derivatives of the KL divergence and setting them equal to zero, we obtain the following pair of update equations:

$$\phi_{ni} \propto \beta_{iw_n} \exp\{\mathrm{E}_q[\log(\theta_i) \mid \gamma]\} \tag{6}$$

$$\gamma_i = \alpha_i + \textstyle\sum_{n=1}^{N} \phi_{ni}. \tag{7}$$

As we show in Appendix A.1, the expectation in the multinomial update can be computed as follows:

$$\mathrm{E}_q[\log(\theta_i) \mid \gamma] = \Psi(\gamma_i) - \Psi\left(\textstyle\sum_{j=1}^{k} \gamma_j\right), \tag{8}$$

where Ψ is the first derivative of the $\log\Gamma$ function which is computable via Taylor approximations (Abramowitz and Stegun, 1970).

Eqs. (6) and (7) have an appealing intuitive interpretation. The Dirichlet update is a posterior Dirichlet given expected observations taken under the variational distribution, $\mathrm{E}[z_n \mid \phi_n]$. The multinomial update is akin to using Bayes' theorem, $p(z_n \mid w_n) \propto p(w_n \mid z_n) p(z_n)$, where $p(z_n)$ is approximated by the exponential of the expected value of its logarithm under the variational distribution.

It is important to note that the variational distribution is actually a conditional distribution, varying as a function of $\mathbf{w}$. This occurs because the optimization problem in Eq. (5) is conducted for fixed $\mathbf{w}$, and thus yields optimizing parameters (γ^*, ϕ^*) that are a function of $\mathbf{w}$. We can write the resulting variational distribution as $q(\theta, \mathbf{z} \mid \gamma^*(\mathbf{w}), \phi^*(\mathbf{w}))$, where we have made the dependence on $\mathbf{w}$ explicit. Thus the variational distribution can be viewed as an approximation to the posterior distribution $p(\theta, \mathbf{z} \mid \mathbf{w}, \alpha, \beta)$.

In the language of text, the optimizing parameters $(\gamma^*(\mathbf{w}), \phi^*(\mathbf{w}))$ are document-specific. In particular, we view the Dirichlet parameters $\gamma^*(\mathbf{w})$ as providing a representation of a document in the topic simplex.